MA1110: Calculus I
Assignment 2 — Full Solutions

Continuity and Differentiability

Note on conventions. Theorem numbers cited below (e.g. Theorem 7.2, Theorem 6.4) refer to the numbering in the course notes *Differentiability* (Ch. 7) and *Continuity* (Ch. 4–6). The exponential and logarithm functions, together with their basic differentiation rules, are treated as already established in the Differentiability notes (see Example 7.1 and Example 7.7 there), so they are used freely in this assignment.

Problem 1

Let $f : \mathbb{R} \to \mathbb{R}$ be additive and continuous, i.e. $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$.

Claim: (B) and (D) must be true; (A) and (C) are false.

(D) holds for every additive function, continuity not even needed. Put $x = y = 0$: $f(0) = 2f(0) \Rightarrow f(0) = 0$. Setting $y = x$ repeatedly gives $f(nx) = nf(x)$ for all $n \in \mathbb{N}$ by induction, and $f(-x) = -f(x)$ since $0 = f(0) = f(x) + f(-x)$. Hence $f(nx) = nf(x)$ for all $n \in \mathbb{Z}$. If $x = p/q \in \mathbb{Q}$ ($q \in \mathbb{N}$), then $qf(x) = f(qx) = f(p \cdot 1) = pf(1)$, so $f(p/q) = \frac{p}{q} f(1)$. Thus with $\lambda = f(1)$, $f(x) = \lambda x$ for every $x \in \mathbb{Q}$. This proves (D).

(B): continuity extends this to all reals. Let $x \in \mathbb{R}$ be arbitrary and choose a sequence of rationals $q_n \to x$ (density of $\mathbb{Q}$). Since f is continuous, $f(q_n) \to f(x)$ (sequential characterization of continuity). But $f(q_n) = \lambda q_n \to \lambda x$. By uniqueness of limits, $f(x) = \lambda x$. As x was arbitrary, $f(x) = \lambda x$ for all $x \in \mathbb{R}$, proving (B).

(A), (C) are false: $f(x) = x^2$ is not additive ($(x + y)^2 \neq x^2 + y^2$ in general), and $f(x) = |x|$ is not additive ($|1 + (-1)| = 0 \neq |1| + |-1| = 2$). Neither can be the f in question.

Problem 2

Let $f : \mathbb{R} \to \mathbb{R}$ be continuous with $f(x) = 0$ for all $x \in \mathbb{Q}$. We show $f \equiv 0$ on $\mathbb{R}$.

Let $x \in \mathbb{R}$ be arbitrary. By density of $\mathbb{Q}$ in $\mathbb{R}$, choose a sequence (q_n) of rationals with $q_n \to x$. By the Sequential Characterization of Continuity, $f(q_n) \to f(x)$. But $f(q_n) = 0$ for every n, so the sequence $(f(q_n))$ is the constant sequence 0, which converges to 0. By uniqueness of limits of sequences, $f(x) = 0$. Since x was arbitrary, $f(x) = 0$ for all $x \in \mathbb{R}$. ■

Problem 3

We want $f : \mathbb{R} \to \mathbb{R}$ with $|f|$ continuous but f discontinuous at every point.

Example 1. Let

$$f_1(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ -1, & x \notin \mathbb{Q}. \end{cases}$$

Then $|f_1(x)| = 1$ for every x, so $|f_1|$ is the constant function 1, which is continuous. To see f_1 is discontinuous at every $a \in \mathbb{R}$: by density of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$, pick a sequence of rationals $q_n \to a$ and a sequence of irrationals $y_n \to a$. Then $f_1(q_n) = 1 \to 1$ while $f_1(y_n) = -1 \to -1$. Since these differ, $\lim_{x\to a} f_1(x)$ does not exist, so f_1 is discontinuous at a. As a was arbitrary, f_1 is nowhere continuous.

Example 2. Let A denote the set of algebraic real numbers (roots of nonzero polynomials with integer coefficients) and let $T = \mathbb{R} \setminus A$ be the transcendental numbers. Both A and T are dense in $\mathbb{R}$ (e.g. $\mathbb{Q} \subset A$ is already dense, and T is dense since it is uncountable while any interval minus A is nonempty). Define

$$f_2(x) = \begin{cases} 1, & x \in A \\ -1, & x \in T. \end{cases}$$

By the identical argument as above (using density of A and T instead of $\mathbb{Q}$ and its complement), $|f_2| \equiv 1$ is continuous, while f_2 is discontinuous at every point of $\mathbb{R}$.

Problem 4

Does $\lim\limits_{h\to 0}[f(x+h) - f(x-h)] = 0$ for every $x \in \mathbb{R}$ imply f is continuous?

No. Counterexample:

$$f(x) = \begin{cases} 0, & x \neq 0 \\ 1, & x = 0. \end{cases}$$

Verification of the hypothesis at every x: Fix $x \neq 0$. For $|h|$ small enough that $0 < |h| < |x|$, both $x + h \neq 0$ and $x - h \neq 0$, so $f(x+h) = f(x-h) = 0$ and the difference is identically 0 for small h; hence the limit as $h \to 0$ is 0.

Fix $x = 0$. For any $h \neq 0$, both $h \neq 0$ and $-h \neq 0$, so $f(h) = f(-h) = 0$, and $f(h) - f(-h) = 0$ for all $h \neq 0$. Hence the limit as $h \to 0$ is 0 as well.

So the hypothesis holds for every $x \in \mathbb{R}$.

But f is discontinuous at 0: take $x_n = 1/n \to 0$. Then $f(x_n) = 0 \to 0 \neq 1 = f(0)$. So f is not continuous at 0. This shows the implication fails. ■

Problem 5

Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(x+y) = f(x)f(y)$ for all x, y, and suppose f is continuous at 0. Show f is continuous on $\mathbb{R}$.

First, $f(0) = f(0+0) = f(0)^2$, so $f(0) \in \{0, 1\}$.

Case $f(0) = 0$. For every $x \in \mathbb{R}$, $f(x) = f(x+0) = f(x)f(0) = 0$. So $f \equiv 0$, which is (trivially) continuous everywhere.

Case $f(0) = 1$. Fix $x \in \mathbb{R}$. We compute, for $h \in \mathbb{R}$,

$$f(x+h) = f(x)f(h).$$

Since f is continuous at 0, $\lim_{h\to 0} f(h) = f(0) = 1$. By the Algebra of Finite Limits,

$$\lim_{h\to 0} f(x+h) = f(x) \lim_{h\to 0} f(h) = f(x) \cdot 1 = f(x).$$

Equivalently, $\lim_{u\to x} f(u) = f(x)$ (substituting $u = x + h$), so f is continuous at x. Since $x \in \mathbb{R}$ was arbitrary, f is continuous on $\mathbb{R}$. ■

Problem 6

$$f(x) = \begin{cases} x, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Continuity only at 0**.**

Continuous at 0*:* For every $x \in \mathbb{R}$, $f(x)$ equals either x or 0, so in either case $|f(x) - f(0)| = |f(x)| \le |x|$. Given $\epsilon > 0$, choose $\delta = \epsilon$; then $|x - 0| < \delta \Rightarrow |f(x) - f(0)| \le |x| < \delta = \epsilon$. So f is continuous at 0.

Discontinuous at every $a \neq 0$*:* Suppose $a \in \mathbb{Q}$, $a \neq 0$, so $f(a) = a \neq 0$. Choose a sequence of irrationals $y_n \to a$ (density of irrationals). Then $f(y_n) = 0 \to 0 \neq a = f(a)$, so f is discontinuous at a.

Suppose $a \notin \mathbb{Q}$, so $f(a) = 0$. Choose a sequence of rationals $x_n \to a$ with $x_n \neq 0$ eventually (possible since $a \neq 0$). Then $f(x_n) = x_n \to a \neq 0 = f(a)$, so f is discontinuous at a.

Hence f is continuous exactly at $x = 0$.

Differentiable nowhere. At any $a \neq 0$, f is not even continuous there, so by Theorem 7.2 (differentiability implies continuity), f cannot be differentiable at a.

At $a = 0$: consider the difference quotient $\dfrac{f(x) - f(0)}{x - 0} = \dfrac{f(x)}{x}$. For $x \in \mathbb{Q} \setminus \{0\}$, $\dfrac{f(x)}{x} = \dfrac{x}{x} = 1$. For $x \notin \mathbb{Q}$, $\dfrac{f(x)}{x} = \dfrac{0}{x} = 0$. Take a sequence of nonzero rationals $x_n \to 0$: the quotient is identically 1. Take a sequence of irrationals $y_n \to 0$: the quotient is identically 0. By the Sequential Characterization of Limits, since these two sequences give different values ($1 \neq 0$), $\lim_{x\to 0} \dfrac{f(x) - f(0)}{x}$ does not exist. So f is not differentiable at 0 either.

Hence f is differentiable at no point of $\mathbb{R}$. ■

Problem 7

$$f_{\alpha,\beta}(x) = \begin{cases} x^\alpha \sin\left(\dfrac{1}{x^\beta}\right), & x > 0 \\ 0, & x = 0. \end{cases}$$

Throughout we use $|\sin(\cdot)| \le 1$, so for $x > 0$,

$$|f_{\alpha,\beta}(x)| \le x^\alpha.$$

We treat the standard case $\beta > 0$ (the case of primary interest, since this is what makes the argument $1/x^\beta$ blow up as $x \to 0^+$ and creates genuine oscillation; for $\beta \le 0$ the argument $1/x^\beta$ stays bounded or tends to 0 and the analysis is easier/degenerate).

(a) Continuity at 0**.**

If $\alpha > 0$: by $(*)$, $|f_{\alpha,\beta}(x)| \le x^\alpha \to 0$ as $x \to 0^+$, so by the Squeeze Theorem $f_{\alpha,\beta}(x) \to 0 = f_{\alpha,\beta}(0)$. Continuous, for every β.

If $\alpha \le 0$ (and $\beta > 0$): choose $x_n = \left(\dfrac{\pi}{2} + 2n\pi\right)^{-1/\beta} \to 0^+$, so that $\sin(1/x_n^\beta) = 1$ for every n. Then $f_{\alpha,\beta}(x_n) = x_n^\alpha$. If $\alpha < 0$, $x_n^\alpha \to \infty$; if $\alpha = 0$, $x_n^\alpha = 1$ for all n. In either case $f_{\alpha,\beta}(x_n) \not\to 0$, so $f_{\alpha,\beta}$ is discontinuous at 0.

Conclusion: $f_{\alpha,\beta}$ **is continuous at** 0 **if and only if** $\alpha > 0$**.**

(b) Differentiability at 0**.**

$$\frac{f_{\alpha,\beta}(x) - f_{\alpha,\beta}(0)}{x - 0} = x^{\alpha-1} \sin\left(\frac{1}{x^\beta}\right), \qquad x > 0.$$

If $\alpha > 1$: $|x^{\alpha-1}\sin(1/x^\beta)| \le x^{\alpha-1} \to 0$ as $x \to 0^+$ (since $\alpha - 1 > 0$), so by the Squeeze Theorem the difference quotient $\to 0$. Hence $f_{\alpha,\beta}$ is differentiable at 0 with $f'_{\alpha,\beta}(0) = 0$.

If $\alpha \le 1$: with x_n as above ($\sin(1/x_n^\beta) = 1$), the quotient equals $x_n^{\alpha-1}$, and $\alpha - 1 \le 0$ forces $x_n^{\alpha-1} \to \infty$ (if $\alpha < 1$) or $x_n^{\alpha-1} = 1$ constant (if $\alpha = 1$, giving oscillation via a second sequence where $\sin = 0$ instead, e.g. $x'_n = (n\pi)^{-1/\beta}$, giving quotient $= 0$); either way the limit fails to exist or diverges.

Conclusion: $f_{\alpha,\beta}$ **is differentiable at** 0 **if and only if** $\alpha > 1$**, and then** $f'_{\alpha,\beta}(0) = 0$**.**

(c) Continuity of $f'_{\alpha,\beta}$ **at** 0**.** (Assume $\alpha > 1$ so that $f'_{\alpha,\beta}(0) = 0$ exists, by (b).) For $x > 0$, by the product and chain rules,

$$f'_{\alpha,\beta}(x) = \alpha x^{\alpha-1} \sin\left(\frac{1}{x^\beta}\right) - \beta x^{\alpha-\beta-1} \cos\left(\frac{1}{x^\beta}\right).$$

The first term $\to 0$ as $x \to 0^+$ since $\alpha - 1 > 0$ (Squeeze Theorem, as in (b)).

For the second term: if $\alpha - \beta - 1 > 0$, i.e. $\alpha > \beta + 1$, then $|-\beta x^{\alpha-\beta-1}\cos(1/x^\beta)| \le |\beta| x^{\alpha-\beta-1} \to 0$, so this term also vanishes in the limit, giving $f'_{\alpha,\beta}(x) \to 0 = f'_{\alpha,\beta}(0)$: continuous.

If $\alpha - \beta - 1 \le 0$, choosing $x_n \to 0^+$ with $1/x_n^\beta = 2n\pi$ (so $\cos = 1$) gives the second term $= -\beta x_n^{\alpha-\beta-1}$, which diverges (if $\alpha < \beta + 1$) or is the nonzero constant $-\beta$ (if $\alpha = \beta + 1$); in neither case does the full expression tend to 0. So $f'_{\alpha,\beta}$ is discontinuous at 0.

Conclusion: $f'_{\alpha,\beta}$ **is continuous at** 0 **if and only if** $\alpha > \beta + 1$**.**

Problem 8

Let f be differentiable on (a, b).

(a) *If $f'(x) \geq 0$ for all $x \in (a, b)$, then f is monotonically increasing.* Let $x_1 < x_2$ in (a, b). Since f is differentiable (hence continuous) on $[x_1, x_2] \subset (a, b)$ and differentiable on (x_1, x_2), the Mean Value Theorem gives $\xi \in (x_1, x_2)$ with

$$f(x_2) - f(x_1) = f'(\xi)(x_2 - x_1).$$

Since $f'(\xi) \geq 0$ and $x_2 - x_1 > 0$, $f(x_2) - f(x_1) \geq 0$, i.e. $f(x_2) \geq f(x_1)$. As $x_1 < x_2$ were arbitrary in (a, b), f is monotonically increasing.

(b) *If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant.* For any $x_1, x_2 \in (a, b)$, the MVT gives ξ between them with $f(x_2) - f(x_1) = f'(\xi)(x_2 - x_1) = 0$, so $f(x_2) = f(x_1)$. Since x_1, x_2 are arbitrary, f is constant on (a, b). (This is the Constant Function Theorem, Theorem 7.14, adapted to an open interval.)

(c) *If $f'(x) \leq 0$ for all $x \in (a, b)$, then f is monotonically decreasing.* Identical to (a): for $x_1 < x_2$, MVT gives $f(x_2) - f(x_1) = f'(\xi)(x_2 - x_1) \leq 0$ since $f'(\xi) \leq 0$, so $f(x_2) \leq f(x_1)$. ■

Problem 9

(a) If f is continuous on $[a, b]$ but fails to be differentiable at even one point of (a, b), the hypotheses of the MVT are not satisfied, so the theorem cannot be invoked to guarantee a point c. However, the *conclusion* of the MVT may still happen to hold for some other reason. Example: $f(x) = x^3$ on $[-1, 1]$ is differentiable everywhere, and even though this doesn't illustrate a failure of differentiability, a better illustrating example is $f(x) = |x| + |x - 1|$ on $[-1, 2]$: f is continuous but not differentiable at $x = 0, 1 \in (-1, 2)$, yet $f'(x) = 0$ on $(0, 1)$ so the conclusion $f'(\xi) = \frac{f(2)-f(-1)}{2-(-1)}$ can still be checked directly: $f(-1) = 1 + 2 = 3$? — the key point is simply: MVT cannot be *applied* (hypothesis violated), though the conclusion sometimes still holds by coincidence.

(b) $f(x) = |x|$ on $[-1, 1]$: MVT fails to apply because f is not differentiable at $x = 0 \in (-1, 1)$ — the differentiability-on-the-open-interval hypothesis is violated. Concretely, no $c \in (-1, 1)$ satisfies $f'(c) = \frac{f(1)-f(-1)}{1-(-1)} = \frac{1-1}{2} = 0$, since $f'(x) = 1$ for $x > 0$ and $f'(x) = -1$ for $x < 0$ (and $f'(0)$ does not exist) — no point gives derivative 0. This confirms the conclusion genuinely fails here.

(c) Yes, c can be unique. If f' is strictly monotonic (in particular, injective) on (a, b), then the equation $f'(x) = \frac{f(b)-f(a)}{b-a}$ has at most one solution in (a, b), so the point c furnished by the MVT is unique. Example: $f(x) = x^2$ on $[a, b]$ with $0 \leq a < b$: here $f'(x) = 2x$ is strictly increasing, and one checks directly $c = \frac{a+b}{2}$ is the unique solution of $2c = \frac{b^2-a^2}{b-a} = a + b$.

Problem 10

Let $f(x) = x^{2026} - x \sin x - \cos x$.

First, f is even: $f(-x) = x^{2026} - (-x)\sin(-x) - \cos(-x) = x^{2026} - (-x)(-\sin x) - \cos x = x^{2026} - x \sin x - \cos x = f(x)$. Also $f(0) = -1 \neq 0$.

The correct answer is (B).

(B) is sufficient. If $f'(x) > 0$ for all $x > 0$, then f is strictly increasing on $(0, \infty)$ (Remark 7.17). Since $f(1) < 0 < f(2)$ and f is continuous and strictly increasing on $[1, 2] \subset (0, \infty)$, the Intermediate Value Theorem gives a zero in $(1, 2)$, and strict monotonicity guarantees this is the *only* zero of f in $(0, \infty)$ (a strictly monotonic function is injective, hence can vanish at most once). Since f is even, this positive zero c has a mirror zero $-c < 0$, and by the same injectivity-via-evenness argument there is no other negative zero either (any negative zero x_0 would force the positive $-x_0$ to be a zero, and we showed there is only one positive zero). Since $f(0) \neq 0$, the total zero count is exactly 2.

Why the others fail. (A) does not control the behavior of f on $(0, 2)$ at all — nothing rules out multiple sign changes there. (C) only tells us f' vanishes once on $(0, \infty)$; f could still be non-monotonic in a way that produces more than one zero (e.g. dip down, come back up, cross again) since we aren't told where the critical point is relative to where $f(1) < 0 < f(2)$, nor anything about behavior outside $[1, 2]$. (D) merely guarantees (via IVT) *at least one* zero in $(1, 2)$ — far from exactly two zeros overall, and doesn't even assert f is even. ■

Problem 11

(a) Suppose f is differentiable on $\mathbb{R}$ with $f'(t) \neq 1$ for every t, and suppose (for contradiction) $x_1 \neq x_2$ are both fixed points, i.e. $f(x_1) = x_1$, $f(x_2) = x_2$. By the MVT applied on the interval between x_1 and x_2, there is c strictly between them with

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{x_2 - x_1}{x_2 - x_1} = 1,$$

contradicting $f'(t) \neq 1$ for all t. Hence f has at most one fixed point.

(b) Let $f(t) = t + (1 + e^t)^{-1}$. Since $1 + e^t > 0$ for all t, $(1 + e^t)^{-1} > 0$ always, so $f(t) - t = (1 + e^t)^{-1} \neq 0$ for every t; hence f has no fixed point.

Now, $f'(t) = 1 - \dfrac{e^t}{(1 + e^t)^2}$. Put $u = e^t > 0$. Since $(1 - u)^2 \geq 0$ we get $(1 + u)^2 \geq 4u$, i.e.

$$0 < \frac{u}{(1 + u)^2} \leq \frac{1}{4}.$$

So $0 \leq \dfrac{e^t}{(1 + e^t)^2} \leq \dfrac{1}{4}$, with the left inequality strict, hence

$$\frac{3}{4} \leq f'(t) < 1,$$

so in particular $0 < f'(t) < 1$ for all $t \in \mathbb{R}$.

(c) Suppose there is $A < 1$ with $|f'(t)| \leq A$ for all $t \in \mathbb{R}$. Let x_1 be arbitrary and $x_{n+1} = f(x_n)$.

Cauchy. By the MVT, for each n, $x_{n+1} - x_n = f(x_n) - f(x_{n-1}) = f'(c_n)(x_n - x_{n-1})$ for some c_n, so $|x_{n+1} - x_n| \leq A|x_n - x_{n-1}|$. By induction, $|x_{n+1} - x_n| \leq A^{n-1}|x_2 - x_1|$. For $m > n$,

$$|x_m - x_n| \leq \sum_{k=n}^{m-1} |x_{k+1} - x_k| \leq |x_2 - x_1| \sum_{k=n}^{m-1} A^{k-1} \leq |x_2 - x_1| \frac{A^{n-1}}{1 - A} \xrightarrow[n \to \infty]{} 0,$$

since $0 \le A < 1$. Hence (x_n) is Cauchy in $\mathbb{R}$, so it converges to some limit x (completeness of $\mathbb{R}$).

x is a fixed point. Since f is differentiable, it is continuous (Theorem 7.2). Taking limits in $x_{n+1} = f(x_n)$ and using continuity,

$$x = \lim_{n\to\infty} x_{n+1} = \lim_{n\to\infty} f(x_n) = f\left(\lim_{n\to\infty} x_n\right) = f(x).$$

So x is a fixed point of f, and by part (a) (since $|f'| \le A < 1$ implies $f'(t) \neq 1$ for all t) it is the unique one. ■

Problem 12

Suppose $C_0 + \frac{C_1}{2} + \cdots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0$. Define

$$F(x) = \sum_{k=0}^{n} \frac{C_k}{k+1} x^{k+1} = C_0 x + \frac{C_1}{2} x^2 + \cdots + \frac{C_n}{n+1} x^{n+1}.$$

F is a polynomial, hence continuous on $[0,1]$ and differentiable on $(0,1)$ (in fact on all of $\mathbb{R}$). Clearly $F(0) = 0$, and $F(1) = \sum_{k=0}^{n} \frac{C_k}{k+1} = 0$ by hypothesis. So $F(0) = F(1)$.

By Rolle's Theorem, there exists $c \in (0,1)$ with $F'(c) = 0$. But

$$F'(x) = \sum_{k=0}^{n} C_k x^k = C_0 + C_1 x + \cdots + C_n x^n.$$

So $C_0 + C_1 c + \cdots + C_n c^n = 0$ for this $c \in (0,1)$, i.e. the given equation has a real root strictly between 0 and 1. ■

Problem 13

Let $f(x) = |x|^3$, i.e. $f(x) = x^3$ for $x \ge 0$ and $f(x) = -x^3$ for $x < 0$.

f': For $x > 0$, $f'(x) = 3x^2$. For $x < 0$, $f'(x) = -3x^2$. At $x = 0$:

$$\frac{f(h) - f(0)}{h} = \frac{|h|^3}{h} = \begin{cases} h^2, & h > 0 \\ -h^2, & h < 0, \end{cases}$$

and both one-sided limits are 0 as $h \to 0$, so $f'(0) = 0$. Combining all cases,

$$f'(x) = 3x|x| \quad \text{for all } x \in \mathbb{R}$$

(check: for $x \ge 0$, $3x|x| = 3x^2$; for $x < 0$, $3x|x| = 3x(-x) = -3x^2$ — consistent).

f'': For $x > 0$, $f''(x) = 6x$. For $x < 0$, $f''(x) = -6x$. At $x = 0$:

$$\frac{f'(h) - f'(0)}{h} = \frac{3h|h|}{h} = 3|h| \to 0,$$

so $f''(0) = 0$. Combining, $f''(x) = 6|x|$ for all $x \in \mathbb{R}$.

$f'''(0)$ **does not exist:**

$$\frac{f''(h)-f''(0)}{h}=\frac{6|h|}{h}=\begin{cases}6, & h>0\\ -6, & h<0.\end{cases}$$

The left-hand limit (-6) and right-hand limit (6) disagree, so $\lim_{h\to 0}\frac{f''(h)-f''(0)}{h}$ does not exist. Hence $f'''(0)$ does not exist. ■

Problem 14

Let J be an interval, $f, g : J \to \mathbb{R}$ differentiable, with $f(a) = f(b) = 0$ for $a, b \in J$. Define

$$h(x) = f(x)e^{g(x)}.$$

Since g is differentiable and e^u is differentiable (Example 7.1/7.7), by the Chain Rule $e^{g(x)}$ is differentiable, and by the Product Rule so is h; in particular h is continuous on $[a, b]$ and differentiable on (a, b).

Also $h(a) = f(a)e^{g(a)} = 0$ and $h(b) = f(b)e^{g(b)} = 0$, so $h(a) = h(b)$. By Rolle's Theorem there is $c \in (a, b)$ with $h'(c) = 0$. Computing,

$$h'(x) = f'(x)e^{g(x)} + f(x)e^{g(x)}g'(x) = e^{g(x)}\big[f'(x) + f(x)g'(x)\big].$$

Since $e^{g(c)} \neq 0$, $h'(c) = 0$ forces

$$f'(c) + f(c)g'(c) = 0. \qquad ■$$

Problem 15

Assume f has a finite derivative on (a, ∞).

(a) Suppose $f(x) \to 1$ and $f'(x) \to c$ as $x \to \infty$. For each n (large), apply the MVT on $[n, n+1]$: there is $\xi_n \in (n, n+1)$ with

$$f(n+1) - f(n) = f'(\xi_n).$$

As $n \to \infty$, $\xi_n \to \infty$ (since $\xi_n > n$), so $f'(\xi_n) \to c$ (as $f'(x) \to c$ as $x \to \infty$, and this holds along every sequence tending to infinity). Also $f(n+1) \to 1$ and $f(n) \to 1$, so the left side $\to 1 - 1 = 0$. Hence $c = 0$.

(b) Suppose $f'(x) \to 1$ as $x \to \infty$; we show $f(x)/x \to 1$. Let $\epsilon > 0$. Choose $M > a$ such that $|f'(t) - 1| < \epsilon$ for all $t > M$. For $x > M$, apply the MVT on $[M, x]$: there is $\xi \in (M, x)$ with

$$f(x) - f(M) = f'(\xi)(x - M), \qquad |f'(\xi) - 1| < \epsilon.$$

Write $f'(\xi) = 1 + \delta$ with $|\delta| < \epsilon$. Then

$$\frac{f(x)}{x} = \frac{f(M)}{x} + \Big(1 - \frac{M}{x}\Big)(1+\delta) = \frac{f(M)}{x} + 1 - \frac{M}{x} + \delta\Big(1 - \frac{M}{x}\Big).$$

So

$$\left|\frac{f(x)}{x} - 1\right| \le \frac{|f(M)|}{x} + \frac{M}{x} + |\delta| \le \frac{|f(M)| + M}{x} + \epsilon.$$

Letting $x \to \infty$ (with M, ϵ fixed), the first term $\to 0$, so $\limsup_{x\to\infty} \left|\frac{f(x)}{x} - 1\right| \le \epsilon$. Since $\epsilon > 0$ was arbitrary, $\displaystyle\lim_{x\to\infty} \frac{f(x)}{x} = 1$.

(c) Suppose $f'(x) \to 0$ as $x \to \infty$; the identical argument (with 1 replaced by 0) gives, for $x > M$,

$$\left|\frac{f(x)}{x}\right| \le \frac{|f(M)|}{x} + \left|1 - \frac{M}{x}\right| |\delta| \le \frac{|f(M)|}{x} + \epsilon,$$

where now $|f'(\xi)| < \epsilon$ so $\delta = f'(\xi)$ itself has $|\delta| < \epsilon$. Letting $x \to \infty$ gives $\limsup |f(x)/x| \le \epsilon$ for every $\epsilon > 0$, so $f(x)/x \to 0$. ■

Problem 16

Suppose f is defined and differentiable for $x > 0$ and $f'(x) \to 0$ as $x \to \infty$. Let $g(x) = f(x+1) - f(x)$. **We prove $g(x) \to 0$ as $x \to \infty$ (the statement is TRUE).**

For each $x > 0$, apply the MVT to f on $[x, x+1]$: there exists $\xi_x \in (x, x+1)$ with

$$f(x+1) - f(x) = f'(\xi_x).$$

As $x \to \infty$, we have $\xi_x > x \to \infty$, so $\xi_x \to \infty$ as well. Since $f'(t) \to 0$ as $t \to \infty$, and $\xi_x \to \infty$, we get $f'(\xi_x) \to 0$. Hence $g(x) = f'(\xi_x) \to 0$ as $x \to \infty$. ■

Problem 17

Suppose $f : \mathbb{R} \to \mathbb{R}$ has both a left derivative and a right derivative at 0 (both finite).

The correct answer is (a): f must be continuous at 0, but may not be differentiable there.

Continuity. Existence of the finite right derivative means $\displaystyle\lim_{h\to 0^+} \frac{f(h) - f(0)}{h}$ exists (finite). If f were not right-continuous at 0, i.e. $\lim_{h\to 0^+} f(h) \ne f(0)$ (or the limit failed to exist), then the numerator $f(h) - f(0)$ would not tend to 0 while the denominator $h \to 0^+$, forcing the quotient to blow up (or fail to converge) — contradicting the existence of a finite right derivative. Hence f must be right-continuous at 0. The identical argument with the left derivative gives left-continuity at 0. Together, f is continuous at 0.

Differentiability can fail. Example: $f(x) = |x|$ has left derivative -1 and right derivative $+1$ at 0 — both exist and are finite, yet they disagree, so f is not differentiable at 0 (Remark 7.3). This shows (c) and (d) are false, and confirms (a) (continuity holds, differentiability need not).

Problem 18

Check differentiability at 0 of:

(A) $f(x) = |x|$**:** Not differentiable at 0 (Remark 7.3: left derivative $-1 \ne 1 =$ right derivative).

(B) $f(x) = |x|^3$**:** Differentiable at 0, with $f'(0) = 0$ (shown in Problem 13).

(C) $f(x) = x^2 \sin(1/x^2)$ for $x \neq 0$, $f(0) = 0$: The difference quotient is

$$\frac{f(h) - f(0)}{h} = h \sin\left(\frac{1}{h^2}\right), \qquad |h \sin(1/h^2)| \leq |h| \to 0,$$

so by the Squeeze Theorem the limit is 0. **Differentiable at** 0, $f'(0) = 0$ (same mechanism as Example 7.4 in the notes).

(D) $f = g^{-1}$ where $g(x) = x^3 + 2x + 3$: $g'(x) = 3x^2 + 2 \geq 2 > 0$ for all x, so g is strictly increasing on $\mathbb{R}$, hence injective; being a continuous odd-degree polynomial it is also onto $\mathbb{R}$ (Example 6.7.3), so g is a bijection and g^{-1} exists on $\mathbb{R}$. Since g' never vanishes, the Inverse Function differentiation rule applies everywhere, so $f = g^{-1}$ is differentiable everywhere, in particular at 0. (Concretely $g(-1) = -1 - 2 + 3 = 0$, so $f(0) = -1$, and $f'(0) = 1/g'(-1) = 1/5$.) **Differentiable at** 0**.**

Summary: (B), (C), (D) are differentiable at 0**; (A) is not.**

Problem 19

(A) False. Take $A = (0, 1)$ and $f(x) = x$. Then f is continuous and bounded above (by 1), and $\sup_{x \in (0,1)} f(x) = 1$, but there is no $c \in (0, 1)$ with $f(c) = 1$. The open interval lacks the compactness needed for the Extreme Value Theorem (Theorem 6.10), which requires a *closed, bounded* interval.

(B) False. Note $A = \bigsqcup_{n=1}^{\infty} [n - \frac{1}{2}, n + \frac{1}{2}] = [\frac{1}{2}, \infty)$, since consecutive intervals share endpoints ($n + \frac{1}{2} = (n + 1) - \frac{1}{2}$). So A is actually one unbounded closed interval. The continuous function $f(x) = x$ on A is unbounded, so continuity on A does not force boundedness (Weierstrass' Theorem, Theorem 6.9, requires a *bounded* closed interval $[a, b]$, which A is not).

(C) True. This is precisely the Intermediate Value Theorem (Theorem 6.4): if f is continuous on $[a, b]$, then f attains every value between $f(a)$ and $f(b)$ at some point of (a, b).

(D) False. Here $A = [1, 2] \cup [3, 4]$ is *disconnected*, so the Fixed Point Theorem (Theorem 6.6, which needs a single interval $[a, b]$) does not apply. Counterexample: define

$$f(x) = \begin{cases} x + 2, & x \in [1, 2] \\ x - 2, & x \in [3, 4]. \end{cases}$$

This is continuous on A (continuous on each closed piece, and the pieces are disjoint so there's no compatibility condition needed at a shared point), and its range is $[3, 4] \cup [1, 2] = A$. But if $c \in [1, 2]$, $f(c) = c \Rightarrow c + 2 = c$, impossible; if $c \in [3, 4]$, $f(c) = c \Rightarrow c - 2 = c$, impossible. So f has no fixed point despite mapping A onto A continuously.

Correct option: (C) only.

Problem 20

(i) $\lim_{x \to 0} x \cos \dfrac{1}{x}$. Since $|x \cos(1/x)| \leq |x| \to 0$, by the Squeeze Theorem the limit **exists and equals** 0.

(ii) $\lim_{x \to 0} \dfrac{x^2 \sin(1/x)}{\sin x}$. Write it as $\left[x \sin \dfrac{1}{x}\right] \cdot \left[\dfrac{x}{\sin x}\right]$. The first bracket $\to 0$ by the Squeeze Theorem ($|x \sin(1/x)| \leq |x| \to 0$). The second bracket $\to 1$ (the standard limit $\sin x / x \to 1$ as $x \to 0$, hence its reciprocal also $\to 1$). A null sequence times a sequence converging to 1 converges to 0. So the limit **exists and equals** 0.

(iii) $\lim_{x\to 0}(1+2x)^{(x+3)/x}$. Write $\dfrac{x+3}{x} = 1 + \dfrac{3}{x}$, so

$$(1+2x)^{1+3/x} = (1+2x)\cdot\left[(1+2x)^{1/x}\right]^3.$$

As $x \to 0$, $(1+2x) \to 1$, and using the standard limit $(1+kx)^{1/x} \to e^k$ (with $k=2$), $(1+2x)^{1/x} \to e^2$, so $\left[(1+2x)^{1/x}\right]^3 \to e^6$. Hence the limit **exists and equals** e^6.

(iv) $\lim_{x\to\infty} x(1+\sin x)$. This limit **does not exist**. Take $x_n = \frac{\pi}{2} + 2n\pi \to \infty$ (so $\sin x_n = 1$): $x_n(1+\sin x_n) = 2x_n \to \infty$. Take $y_n = \frac{3\pi}{2} + 2n\pi \to \infty$ (so $\sin y_n = -1$): $y_n(1+\sin y_n) = 0 \to 0$. Two sequences tending to ∞ produce different limiting behavior (∞ vs. 0) for the expression, so the overall limit does not exist.

Problem 21

Let $f(x) = (x-1)^3 + (x-2)^3 + (x-3)^3 + (x-4)^3$.

At least one real root: f is a polynomial of degree 3 (odd degree — the leading term is $4x^3$), so by Example 6.7.3 (odd-degree polynomials are onto $\mathbb{R}$), f has at least one real root.

At most one real root: We have

$$f'(x) = 3(x-1)^2 + 3(x-2)^2 + 3(x-3)^2 + 3(x-4)^2 = 3\left[(x-1)^2 + (x-2)^2 + (x-3)^2 + (x-4)^2\right].$$

This is a sum of squares, hence ≥ 0; moreover it can equal 0 only if all four squares vanish simultaneously, which would require $x = 1, 2, 3, 4$ all at once — impossible. So $f'(x) > 0$ strictly for every $x \in \mathbb{R}$.

Suppose, for contradiction, $x_1 < x_2$ were two distinct roots. By Rolle's Theorem there would exist $c \in (x_1, x_2)$ with $f'(c) = 0$, contradicting $f' > 0$ everywhere. So f has at most one real root.

Combining, f has *exactly* one real root. ■

Problem 22

Let $f : [0,2] \to \mathbb{R}$ be differentiable on $(0,2)$, with $f(0) = 0, f(1) = 4, f(2) = 2$.

By the MVT on $[0,1]$: there is $c_1 \in (0,1)$ with

$$f'(c_1) = \frac{f(1) - f(0)}{1 - 0} = 4.$$

By the MVT on $[1,2]$: there is $c_2 \in (1,2)$ with

$$f'(c_2) = \frac{f(2) - f(1)}{2 - 1} = 2 - 4 = -2.$$

So $f'(c_1) = 4$ and $f'(c_2) = -2$ are different values taken by the function f' at $c_1 < c_2$, both in $(0,2)$. By Darboux's Theorem (Theorem 7.20 — the derivative of a differentiable function has the intermediate value property, even without assuming f' is continuous), since 0 lies between -2 and 4, there exists c between c_1 and c_2 (hence $c \in (0,2)$) with

$$f'(c) = 0. \qquad \blacksquare$$

Problem 23

Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $|f(x) - f(y)| \le C|x-y|^2$ for all $x, y \in \mathbb{R}$, some constant $C > 0$.

Fix $c \in \mathbb{R}$ and $h \neq 0$. Taking $x = c+h, y = c$:

$$|f(c+h) - f(c)| \le Ch^2 \implies \left|\frac{f(c+h) - f(c)}{h}\right| \le C|h|.$$

As $h \to 0$, the right side $\to 0$, so by the Squeeze Theorem,

$$\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = 0,$$

i.e. $f'(c) = 0$. Since $c \in \mathbb{R}$ was arbitrary, $f'(x) = 0$ for every $x \in \mathbb{R}$. By the Constant Function Theorem (Theorem 7.14), f is constant on $\mathbb{R}$. $\blacksquare$

Problem 24

Let $f(x) = |x| + |x-1| + |x-2|$. We compute f (and f') piecewise.

$$f(x) = \begin{cases} 3 - 3x, & x < 0 \\ 3 - x, & 0 < x < 1 \\ x + 1, & 1 < x < 2 \\ 3x - 3, & x > 2, \end{cases} \qquad f'(x) = \begin{cases} -3, & x < 0 \\ -1, & 0 < x < 1 \\ 1, & 1 < x < 2 \\ 3, & x > 2. \end{cases}$$

At $x = 0$: left derivative $= -3$, right derivative $= -1$; these disagree, so $f'(0)$ does not exist.

At $x = 1$: left derivative $= -1$, right derivative $= 1$; disagree, so $f'(1)$ does not exist.

At $x = 2$: left derivative $= 1$, right derivative $= 3$; disagree, so $f'(2)$ does not exist.

So $f'(x) = -3$ for $x < 0$; -1 for $0 < x < 1$; 1 for $1 < x < 2$; 3 for $x > 2$; and f' does not exist at $x = 0, 1, 2$.

Problem 25

$$f(x) = \begin{cases} x, & 0 \le x \le 1 \\ 2 - x^2, & 1 \le x \le 2 \\ x - x^2, & 2 \le x \le 3. \end{cases}$$

On the open pieces $(0,1), (1,2), (2,3)$, f is a polynomial and hence differentiable there, with $f'(x) = 1$ on $(0,1)$, $f'(x) = -2x$ on $(1,2)$, $f'(x) = 1-2x$ on $(2,3)$.

At $x = 1$: first check continuity — from the left, $f(1^-) = 1$; from the right (using $2-x^2$), value at $x = 1$ is $2-1 = 1$; consistent, $f(1) = 1$, continuous. Left derivative: $\frac{d}{dx}(x) = 1$. Right derivative: $\frac{d}{dx}(2-x^2) = -2x$ at $x = 1$ gives -2. Since $1 \neq -2$, f **is not differentiable at $x = 1$.**

At $x = 2$: continuity — left piece $2-x^2$ at $x = 2$ gives $2-4 = -2$; right piece $x - x^2$ at $x = 2$ gives $2-4 = -2$; consistent, $f(2) = -2$, continuous. Left derivative: $\frac{d}{dx}(2-x^2) = -2x$ at $x = 2$ gives -4. Right derivative: $\frac{d}{dx}(x-x^2) = 1-2x$ at $x = 2$ gives $1-4 = -3$. Since $-4 \neq -3$, f **is not differentiable at $x = 2$.**

At the endpoints $x = 0$ and $x = 3$, only a one-sided derivative can be discussed (the domain doesn't extend further): $f'(0^+) = 1$, and $f'(3^-) = 1 - 2(3) = -5$.

Conclusion: f is differentiable at every point of $[0,3]$ *except* $x = 1$ and $x = 2$, where the left- and right-hand derivatives disagree.

Problem 26

Let $f : [a,b] \to \mathbb{R}$ be continuous, and let $x_1, x_2, x_3 \in [a,b]$ be arbitrary.

By the Extreme Value Theorem (Theorem 6.10), f attains a maximum $M = f(x_{\max})$ and minimum $m = f(x_{\min})$ at some points $x_{\max}, x_{\min} \in [a,b]$, and $m \leq f(x) \leq M$ for every $x \in [a,b]$.

In particular $m \leq f(x_i) \leq M$ for $i = 1,2,3$, so their average also satisfies

$$m \leq \frac{f(x_1) + f(x_2) + f(x_3)}{3} \leq M$$

(an average of three numbers each in $[m, M]$ stays in $[m, M]$).

Since this average lies between $f(x_{\min}) = m$ and $f(x_{\max}) = M$, and f is continuous on the interval between $x_{\min}$ and $x_{\max}$ (a subset of $[a,b]$), the Intermediate Value Theorem (Theorem 6.4) gives a point c between $x_{\min}$ and $x_{\max}$ (hence $c \in [a,b]$) with

$$f(c) = \frac{f(x_1) + f(x_2) + f(x_3)}{3}. \qquad \blacksquare$$

Problem 27

Let $f : [a,b] \to \mathbb{R}$ be continuous with $f(x) > 0$ for all $x \in [a,b]$.

By the Extreme Value Theorem (Theorem 6.10), f attains a minimum value $m = f(x_{\min})$ for some $x_{\min} \in [a,b]$. Since $f(x_{\min}) > 0$ by hypothesis, $m > 0$.

Let $\alpha = \dfrac{m}{2} > 0$. Then for every $x \in [a,b]$,

$$f(x) \geq m > \frac{m}{2} = \alpha,$$

so $f(x) > \alpha$ for all $x \in [a,b]$, as required. $\blacksquare$

Problem 28

Two examples of $f : [a, b] \to \mathbb{R}$ with the intermediate value property (IVP) but not continuous:

Example 1. On $[-1, 1]$, let $f(x) = 2x\sin(1/x) - \cos(1/x)$ for $x \neq 0$, $f(0) = 0$. This is exactly the derivative of $g(x) = x^2 \sin(1/x)$ (with $g(0) = 0$) computed in Example 7.4 of the notes, where it is shown $f = g'$ oscillates and has no limit at 0, so f is discontinuous at 0. However, since f arises as the derivative of a function differentiable everywhere on $[-1, 1]$, Darboux's Theorem (Theorem 7.20) guarantees f has the intermediate value property on $[-1, 1]$ despite this discontinuity.

Example 2. On $[-1, 1]$, let $f(x) = 1 + 2x\sin(1/x) - \cos(1/x)$ for $x \neq 0$, $f(0) = 1$. This is the derivative of $h(x) = x + x^2 \sin(1/x)$ (with $h(0) = 0$), also computed in Example 7.4 of the notes; again f oscillates and is discontinuous at 0, but by Darboux's Theorem it has the intermediate value property on $[-1, 1]$.

(In both cases, being a derivative is what forces the IVP via Darboux's theorem, while the explicit oscillation of $\cos(1/x)$ near 0 is what destroys continuity — this is precisely why these are good illustrating examples.)

Problem 29

Let $f : (-1, 1) \to \mathbb{R}$ be continuous with $f(x) = f(x^2)$ for all $x \in (-1, 1)$.

Fix $x \in (-1, 1)$. Iterating the functional equation,

$$f(x) = f(x^2) = f\big((x^2)^2\big) = f(x^4) = f(x^8) = \cdots = f\big(x^{2^n}\big) \quad \text{for every } n \in \mathbb{N}.$$

Since $|x| < 1$, we have $x^{2^n} \to 0$ as $n \to \infty$. By continuity of f at 0 and the Sequential Characterization of Continuity, since $x^{2^n} \to 0$,

$$f(x^{2^n}) \to f(0).$$

But $f(x^{2^n}) = f(x)$ for every n (a constant sequence), so its limit is trivially $f(x)$. By uniqueness of limits, $f(x) = f(0)$.

Since $x \in (-1, 1)$ was arbitrary, $f(x) = f(0)$ for all x, i.e. f is constant on $(-1, 1)$. ■

Problem 30

Suppose $f'(x)$ exists on $(0, 1)$ (with f continuous on $[0, 1]$, as is implicit). Apply Cauchy's Mean Value Theorem (Theorem 7.19) to f and $g(x) = x^3$ on $[0, 1]$: both are continuous on $[0, 1]$ and differentiable on $(0, 1)$, so there exists $x \in (0, 1)$ with

$$[f(1) - f(0)]\, g'(x) = [g(1) - g(0)]\, f'(x).$$

Since $g(x) = x^3$, $g'(x) = 3x^2$, $g(1) = 1$, $g(0) = 0$, this becomes

$$[f(1) - f(0)] \cdot 3x^2 = [1 - 0] \cdot f'(x) = f'(x),$$

i.e.

$$3x^2[f(1) - f(0)] = f'(x)$$

for this $x \in (0, 1)$. This is exactly the equation in question, so it has at least one solution in $(0, 1)$. ■